

IP Addresses: Classful Addressing

Binary Notation

01110101 10010101 00011101 11101010

Dotted-decimal notation

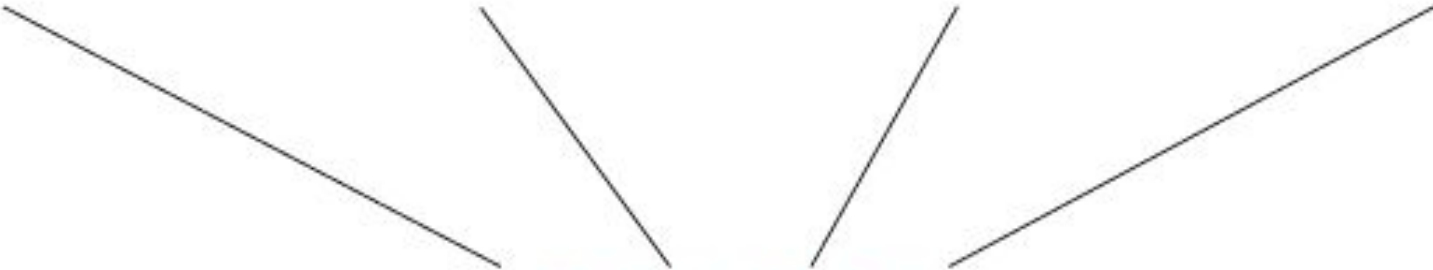
10000000

00001011

00000011

00011111

128.11.3.31



Example 1

Change the following IP address from binary notation to dotted-decimal notation.

10000001 00001011 00001011 11101111

Solution

129.11.11.239

Example 2

Change the following IP address from dotted-decimal notation to binary notation.

111.56.45.78

Solution

01101111 00111000 00101101 01001110

Finding the class in binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

Example 6

Find the class of the address:

00000001 00001011 00001011 11101111

Solution

The first bit is 0. This is a class A address.

Example 6 (Continued)

Find the class of the address:

11000001 10000011 00011011 11111111

Solution

The first 2 bits are 1; the third bit is 0.

This is a class C address.

Finding the class in decimal notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0 to 127			
Class B	128 to 191			
Class C	192 to 223			
Class D	224 to 239			
Class E	240 to 255			

Example 7 (Continued)

Find the class of the address:

193.14.56.22

Solution

The first byte is 193 (between 192 and 223);
the class is C.

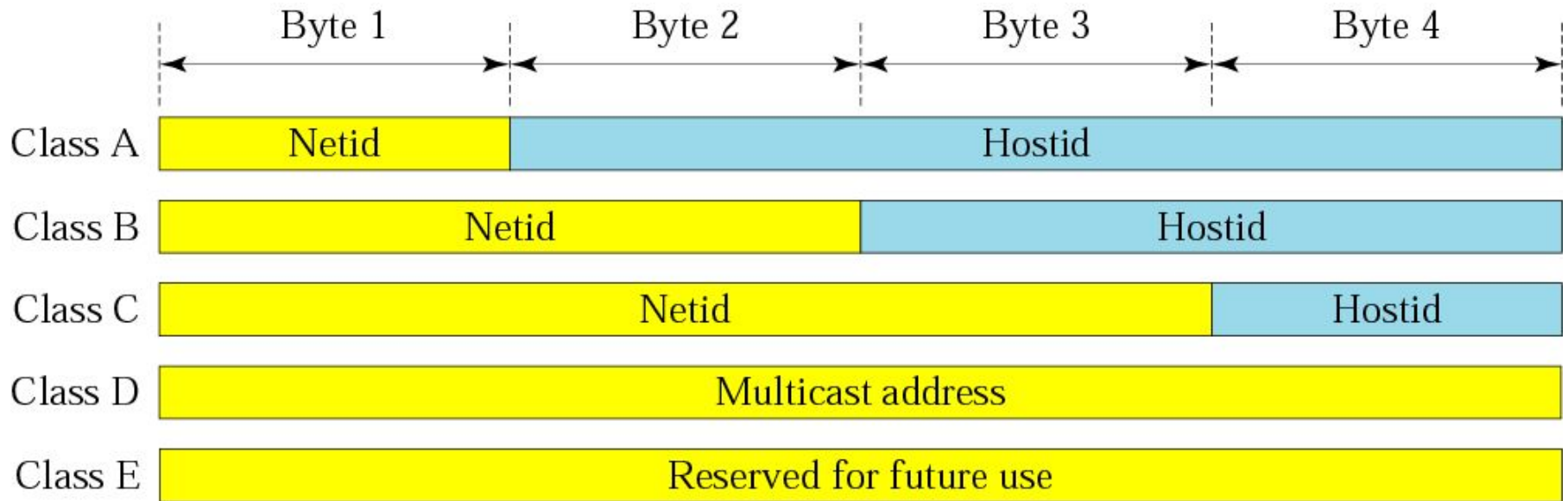
Example 5

How can we prove that we have 2,147,483,648 addresses in class A?

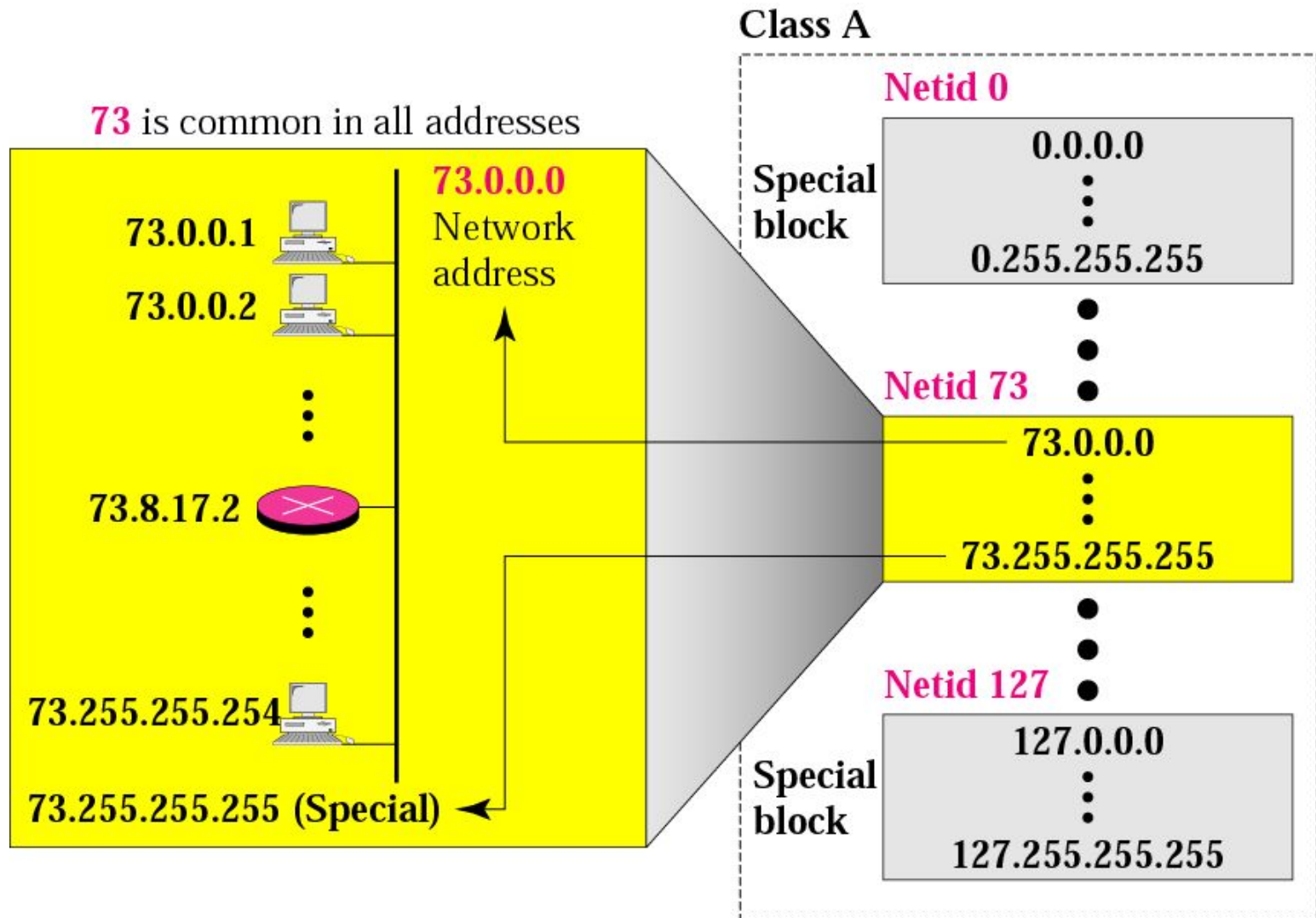
Solution

In class A, only 1 bit defines the class. The remaining 31 bits are available for the address. With 31 bits, we can have 2^{31} or 2,147,483,648 addresses.

Netid and hostid



Blocks in class A

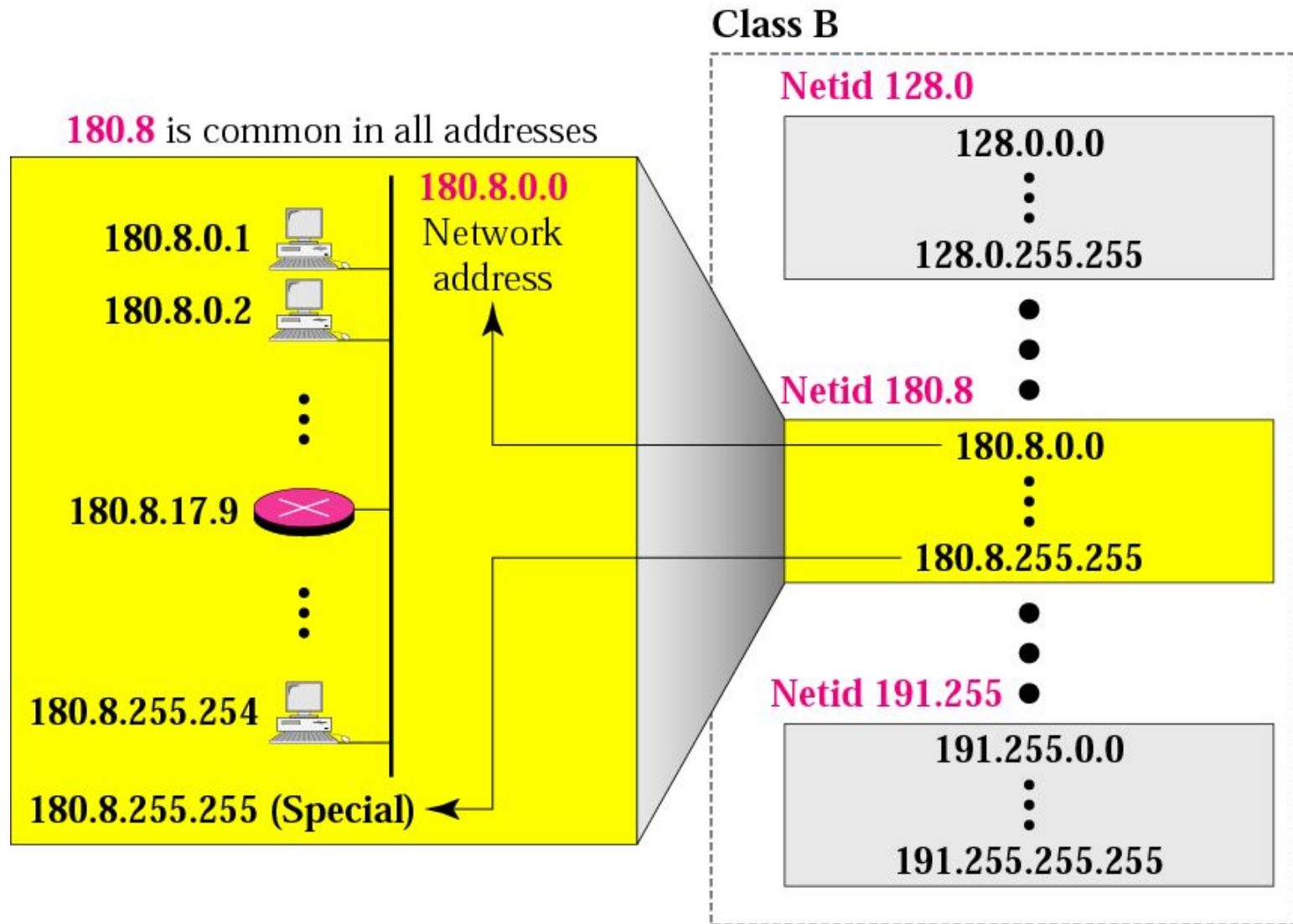


128 blocks: 16,777,216 addresses in each block

Note

*Millions of class A addresses
are wasted.*

Blocks in class B

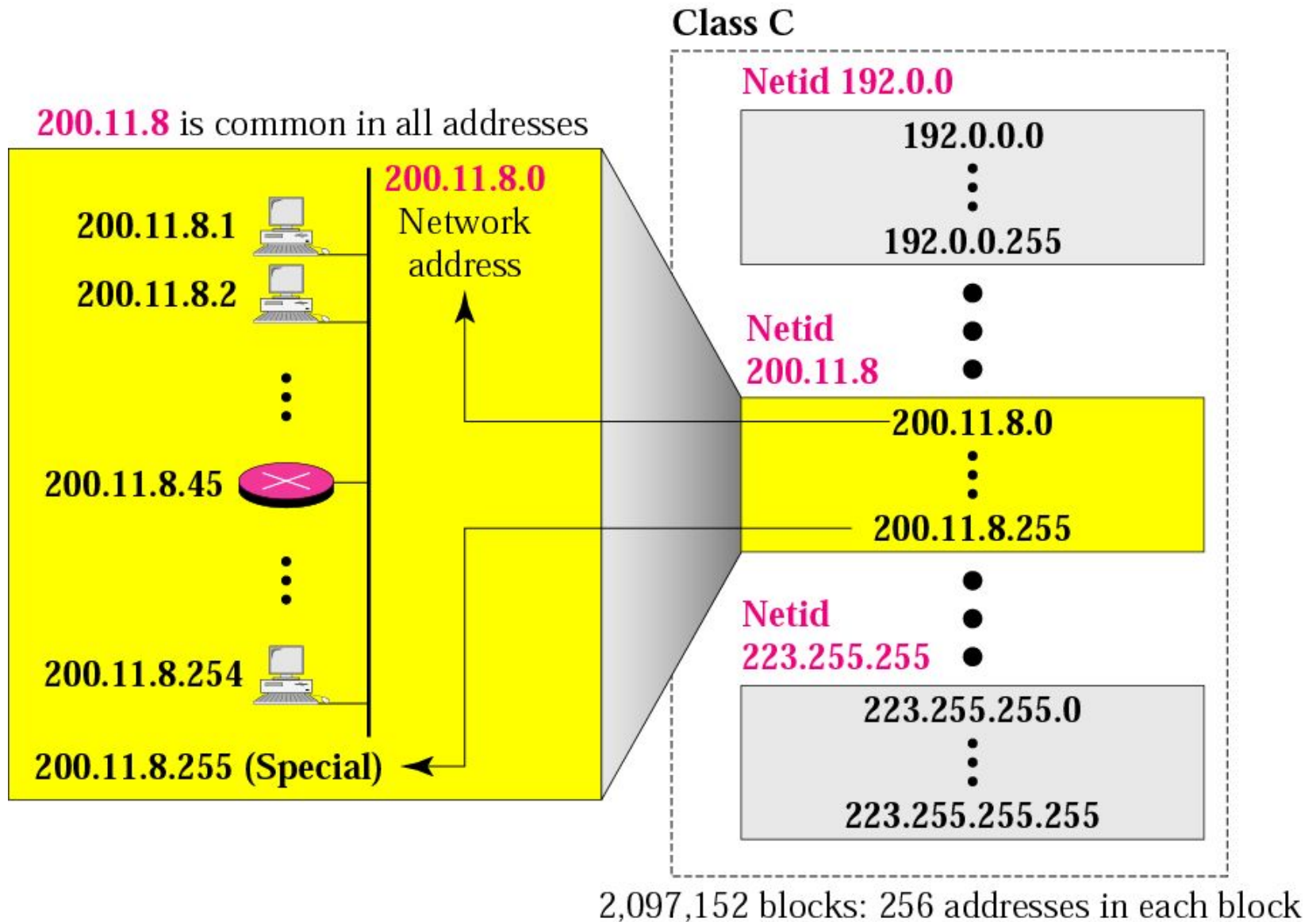


16,384 blocks: 65,536 addresses in each block

Note

*Many class B addresses
are wasted.*

Blocks in class C



Note

*The number of addresses in
a class C block
is smaller than
the needs of most organizations.*

Network Addresses

The network address is the first address.

The network address defines the network to the rest of the Internet.

Given the network address, we can find the class of the address, the block, and the range of the addresses in the block

Note

*In classful addressing,
the network address
(the first address in the block)
is the one that is assigned
to the organization.*

Example 9

Given the network address 17.0.0.0, find the class, the block, and the range of the addresses.

Solution

The class is A because the first byte is between 0 and 127. The block has a netid of 17. The addresses range from 17.0.0.0 to 17.255.255.255.

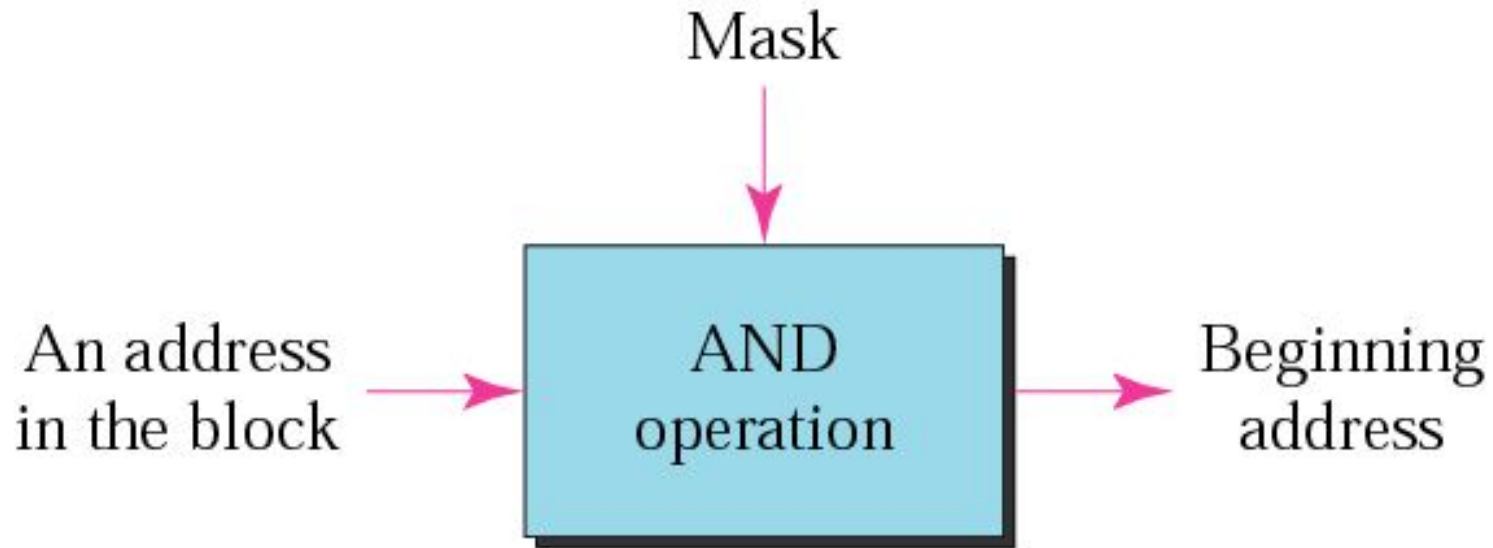
Example 10

Given the network address 132.21.0.0, find the class, the block, and the range of the addresses.

Solution

The class is B because the first byte is between 128 and 191. The block has a netid of 132.21. The addresses range from 132.21.0.0 to 132.21.255.255.

Masking concept



A mask is a 32-bit binary number that gives the first address in the block (the network address) when bitwise ANDed with an address in the block.

Example 12

Given the address 23.56.7.91 and the default class A mask, find the beginning address (network address).

Solution

The default mask is 255.0.0.0, which means

that only the first byte is preserved and the other 3 bytes are set to 0s.

The network address is 23.0.0.0.

Example 13

Given the address 132.6.17.85 and the default class B mask, find the beginning address (network address).

Solution

The default mask is 255.255.0.0, which means that the first 2 bytes are preserved and the other 2 bytes are set to 0s. The network address is 132.6.0.0.

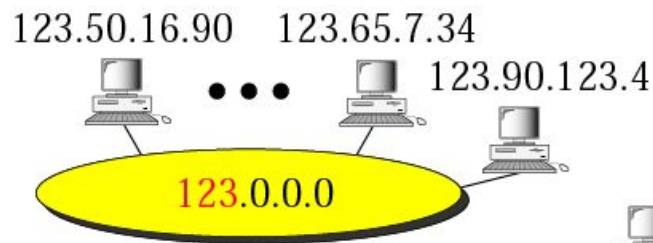
Example 14

Given the address 201.180.56.5 and the class C default mask, find the beginning address (network address).

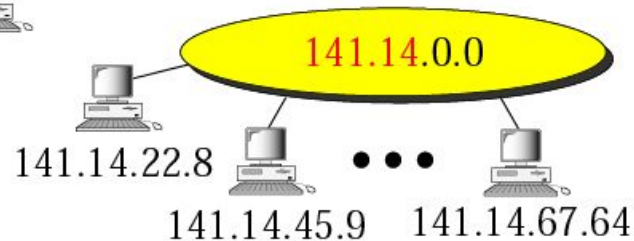
Solution

The default mask is 255.255.255.0, which means that the first 3 bytes are preserved and the last byte is set to 0.

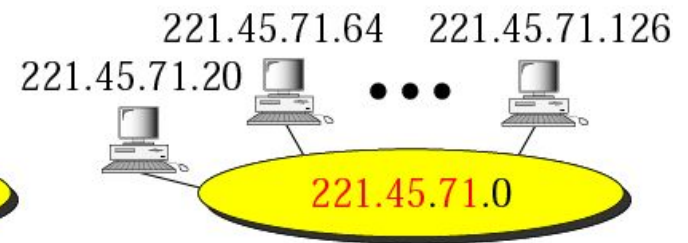
Network addresses



(a) Class A

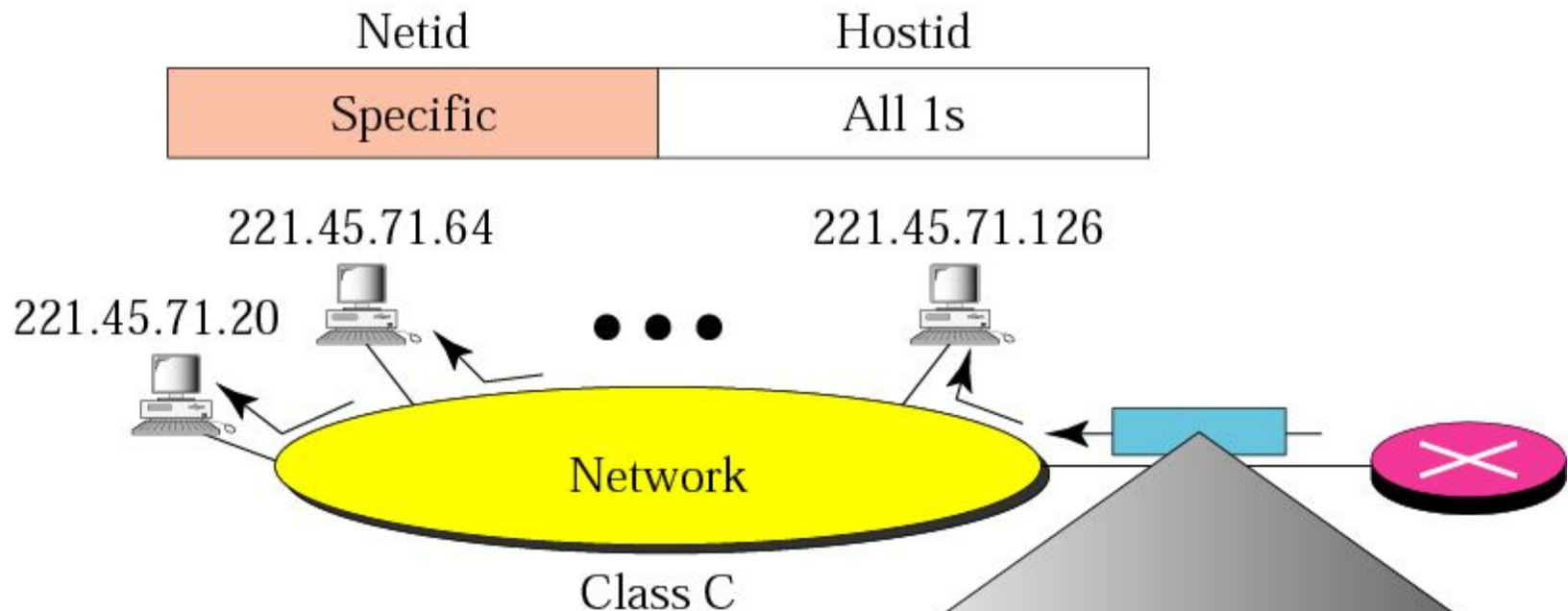


(b) Class B



(c) Class C

Example of direct broadcast address



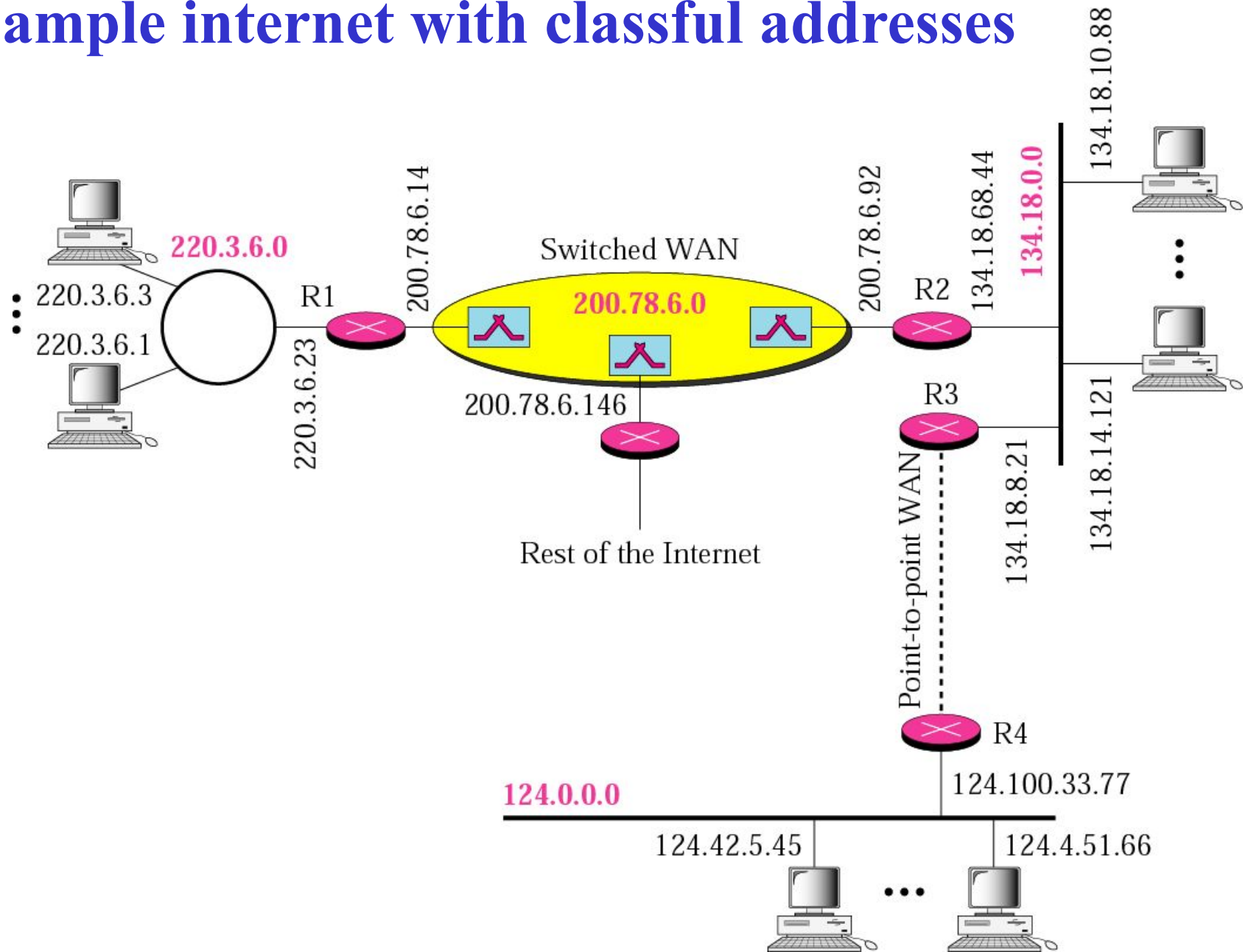
The direct broadcast address is used by a router to send a message to every host on a local network. Every host/router receives and processes the packet with a direct broadcast address.

Destination IP address:
221.45.71.255
Hostid: 255

Private Addresses

A number of blocks in each class are assigned for private use. They are not recognized globally.

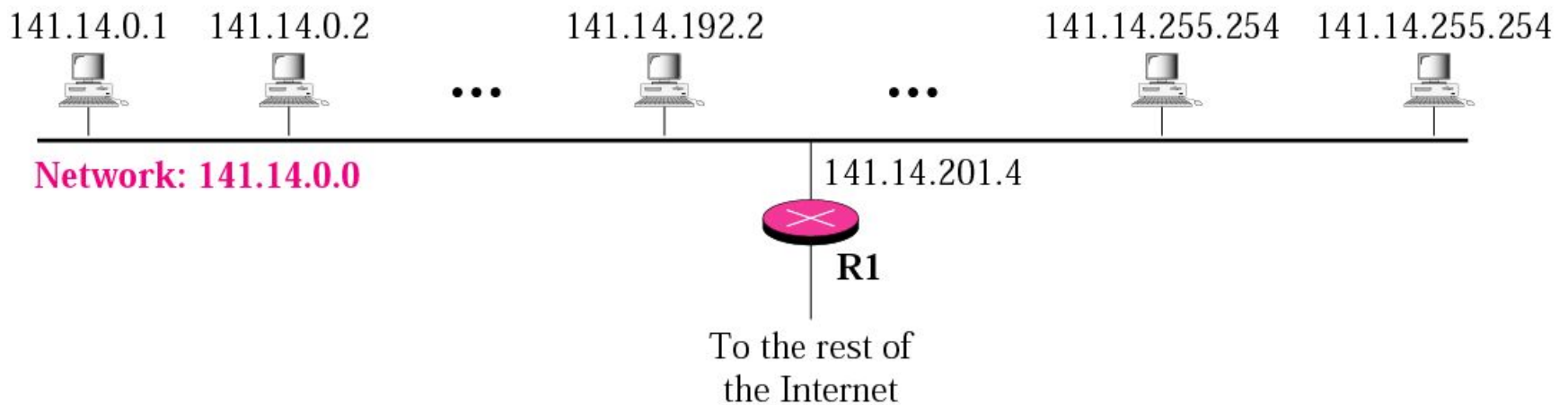
Sample internet with classful addresses



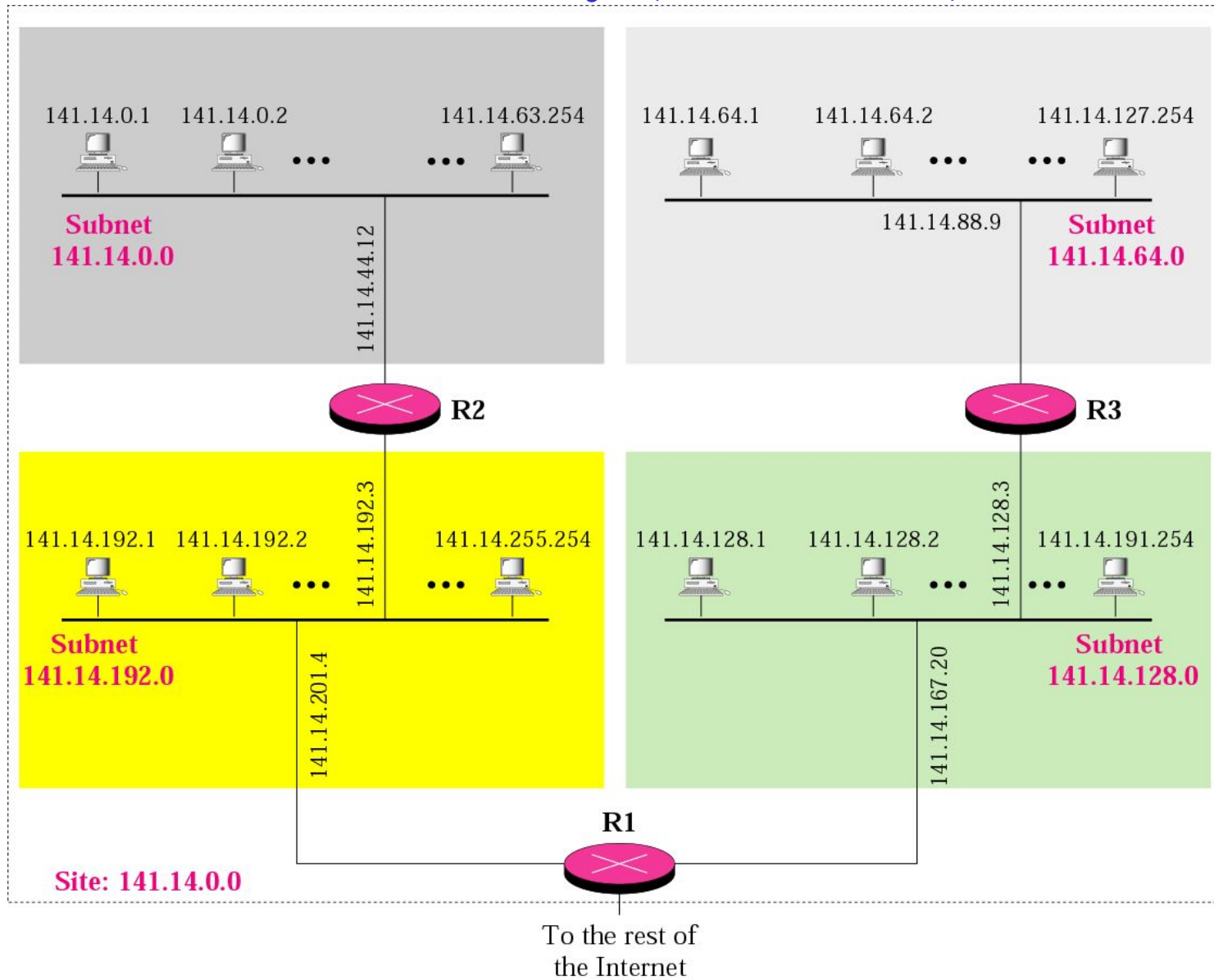
5.1

SUBNETTING

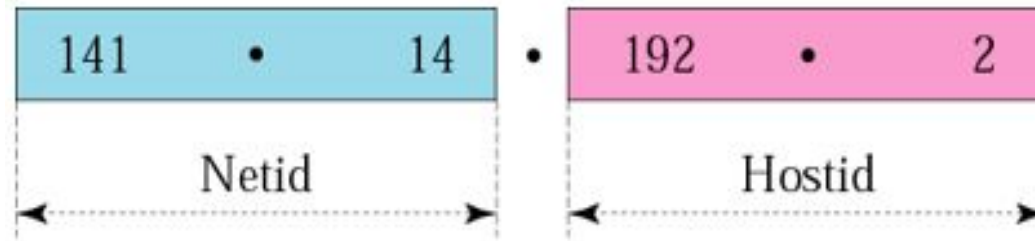
A network with two levels of hierarchy (not subnetted)



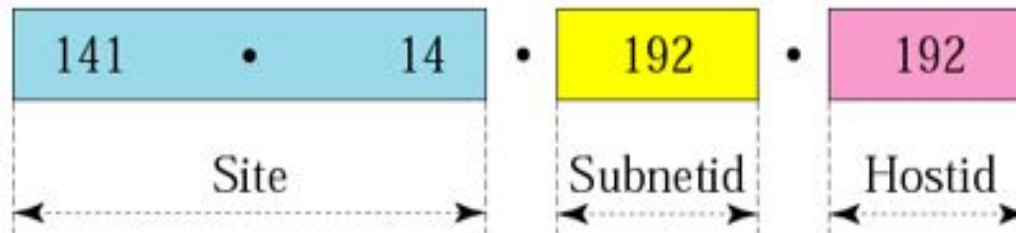
A network with three levels of hierarchy (subnetted)



Addresses in a network with and without subnetting



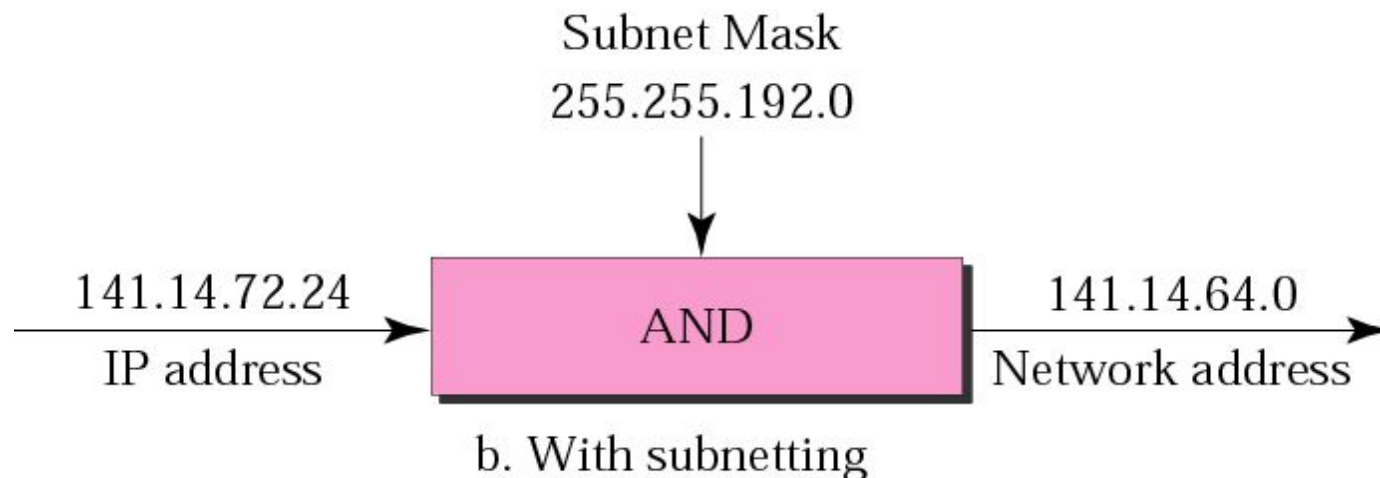
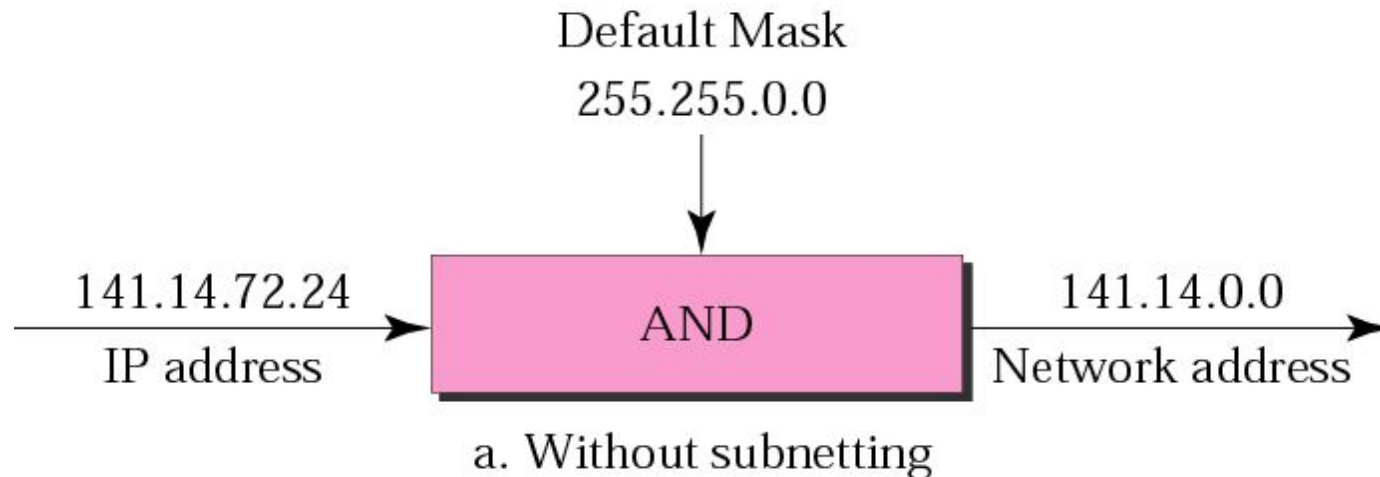
a. Without subnetting



b. With subnetting

Finding the Subnet Address

Given an IP address, we can find the subnet address the same way we found the network address.



Example 1

What is the subnetwork address if the destination address is 146.45.34.56 and the subnet mask is 255.255.240.0?

10010010 00101101 00100010 00111000

11111111 11111111 11110000 00000000

11001000 00101101 00100000 00000000

The subnetwork address is **146.45.32.0**.

Example 2

What is the subnetwork address if the destination address is 19.30.80.5 and the mask is 255.255.192.0?

Solution

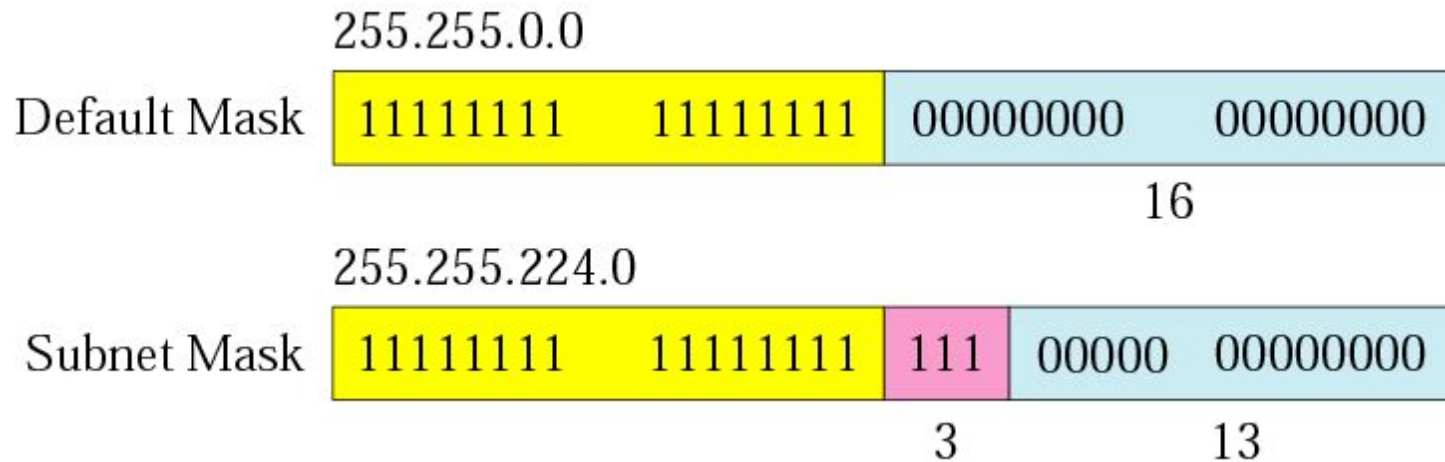
IP Address

19	•	30	•	84	•	5
Mask						
255	•	255	•	192	•	0
19	•	30	•	64	•	0

Subnet Address

84	0	1	0	1	0	1	0	0
192	1	1	0	0	0	0	0	0
64	0	1	0	0	0	0	0	0

Comparison of a default mask and a subnet mask



Note

The number of subnets must be a power of 2.

Example 3

A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets.

Solution

The number of 1s in the default mask is 24 (class C).

The company needs six subnets. This number 6 is not a power of 2. The next number that is a power of 2 is 8 (2^3). We need 3 more 1s in the subnet mask. The total number of 1s in the subnet mask is 27 ($24 + 3$). The total number of 0s is 5 ($32 - 27$). The mask is

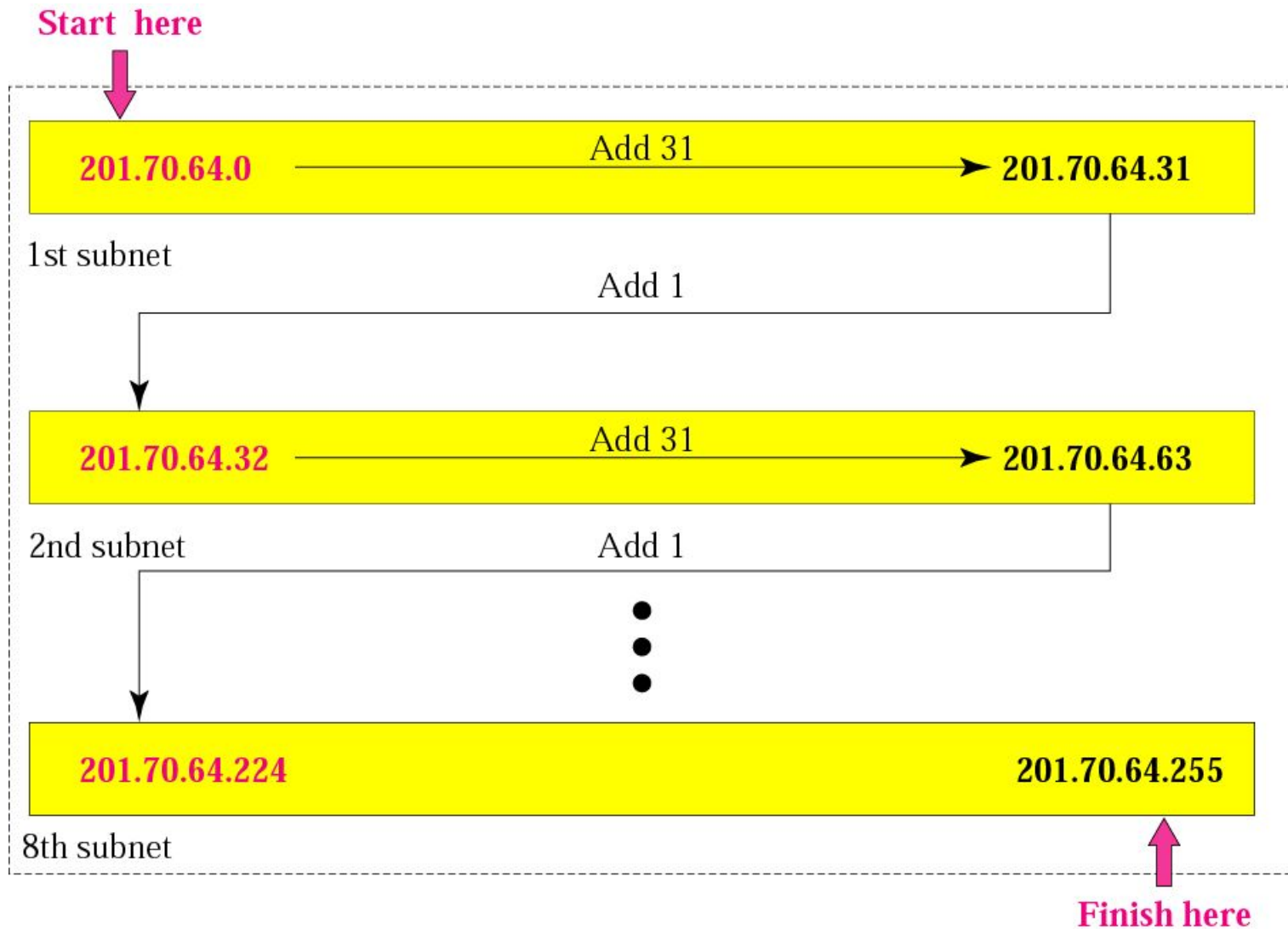
11111111 11111111 11111111 11100000

or

255.255.255.224

Figure 5-8

Example 3



Example 4

A company is granted the site address 181.56.0.0 (class B). The company needs 1000 subnets. Design the subnets.

Solution

The number of 1s in the default mask is 16 (class B). The company needs 1000 subnets. This number is not a power of 2. The next number that is a power of 2 is 1024 (2^{10}). We need 10 more 1s in the subnet mask. The total number of 1s in the subnet mask is 26 ($16 + 10$). The total number of 0s is 6 ($32 - 26$). The mask is

11111111 11111111 11111111 110000000

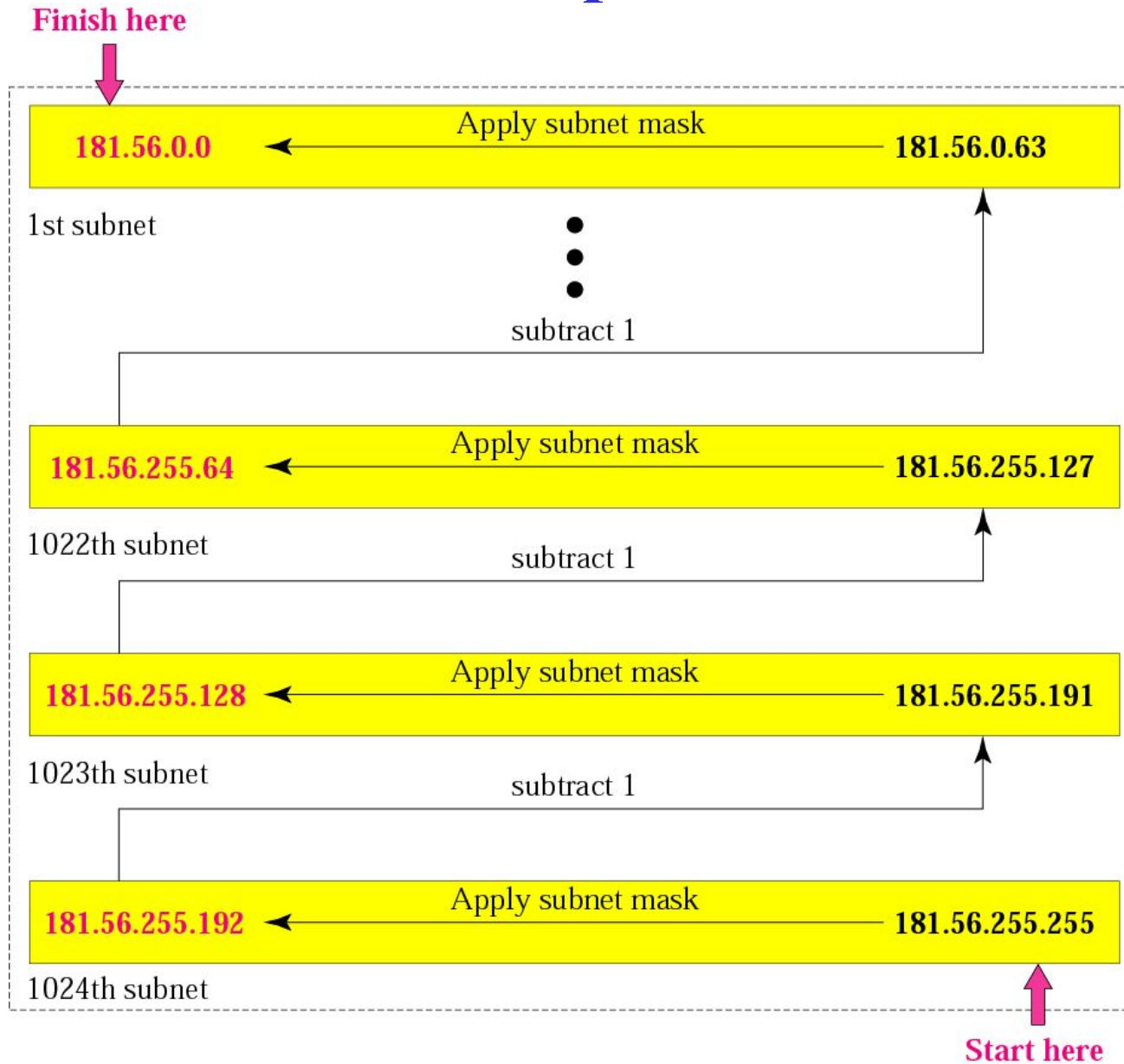
or

255.255.255.192.

The number of subnets is 1024. The number of addresses in each subnet is 2^6 (6 is the number of 0s) or 64.

Figure 5-9

Example 4



5.2

SUPERNETTING

(Self Study)

Figure 5-11

A supernetwork

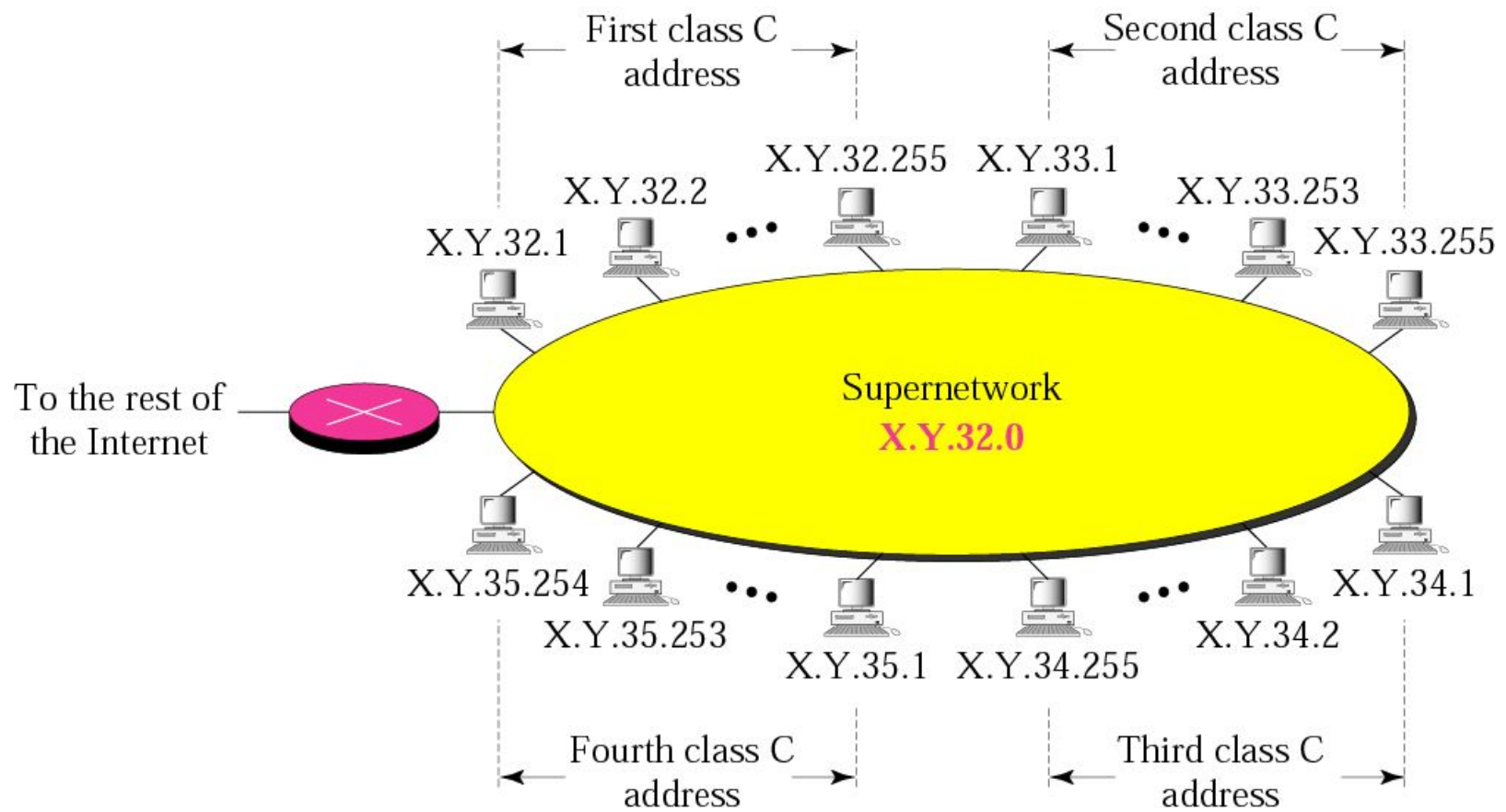
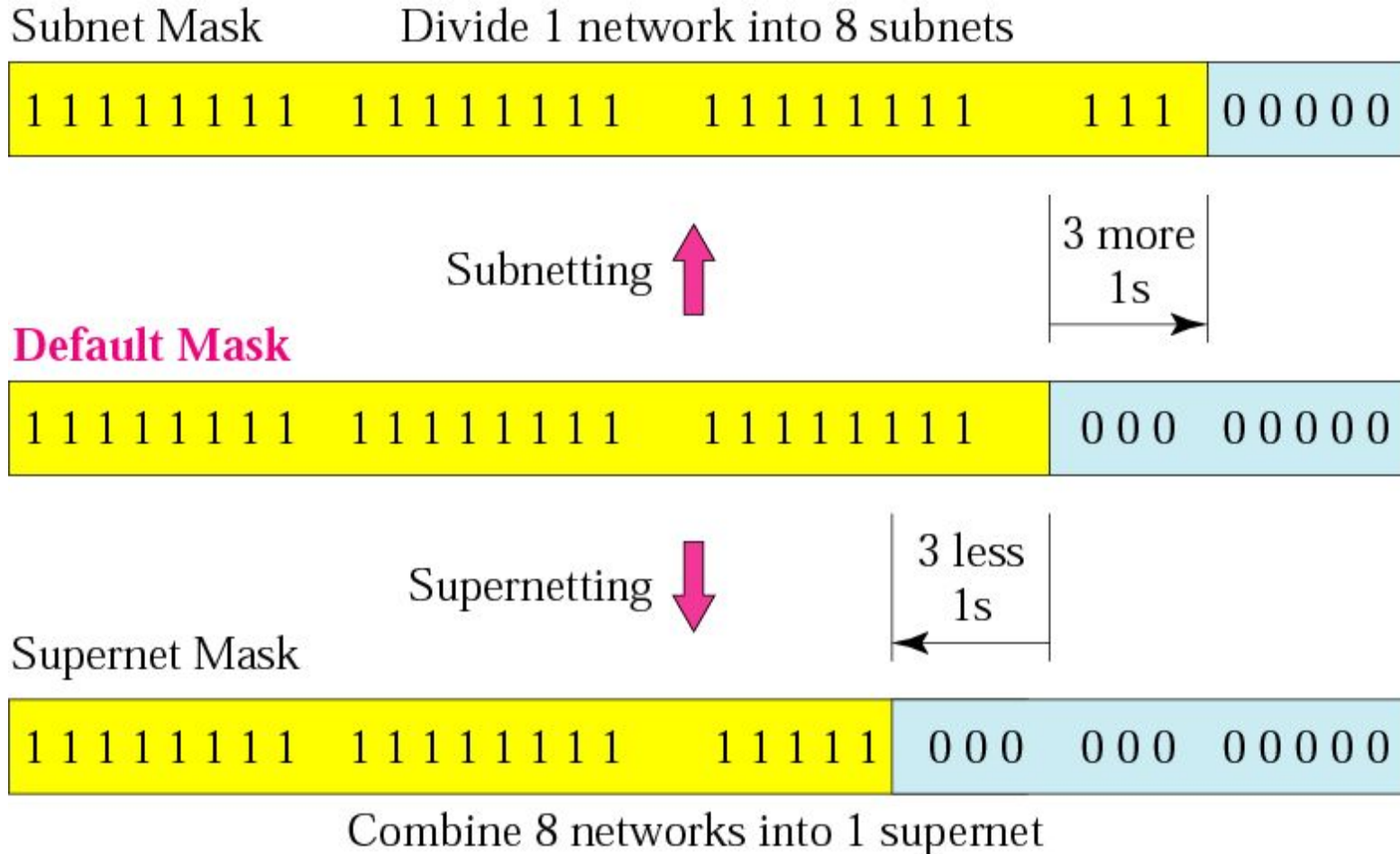


Figure 5-12

Comparison of subnet, default, and supernet masks



Example 6

We need to make a supernet out of 16 class C blocks. What is the supernet mask?

Solution

We need 16 blocks. For 16 blocks we need to change four 1s to 0s in the default mask. So the mask is

11111111 11111111 11110000 00000000

or

255.255.240.0

Example 7

A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. A router receives three packets with the following destination addresses:

205.16.37.44

205.16.42.56

205.17.33.76

Which packet belongs to the supernet?

Solution

We apply the supernet mask to see if we can find the beginning address.

205.16.37.44 AND 255.255.248.0 ☐ 205.16.32.0

205.16.42.56 AND 255.255.248.0 ☐ 205.16.40.0

205.17.33.76 AND 255.255.248.0 ☐ 205.17.32.0

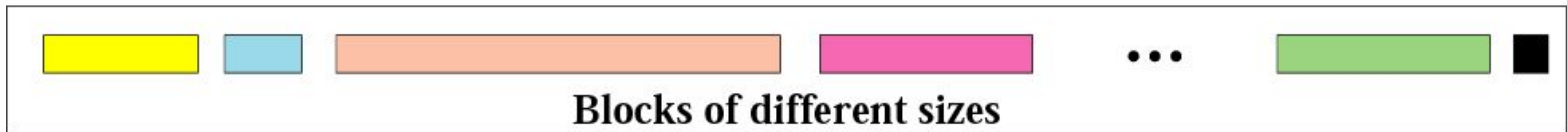
Only the first address belongs to this supernet.

5.3

CLASSLESS ADDRESSING

Variable-length blocks

Address Space



Slash notation

A.B.C.D/*n*

Note

*Slash notation is also called
CIDR
notation.*

Example 11

A small organization is given a block with the beginning address and the prefix length **205.16.37.24/29** (in slash notation). What is the range of the block?

Solution

The beginning address is 205.16.37.24. To find the last address we keep the first 29 bits and change the last 3 bits to 1s.

Beginning: 11001111 00010000 00100101 00011000

Ending : 11001111 00010000 00100101 00011111

There are only 8 addresses in this block.

Example 12

We can find the range of addresses in Example 11 by another method. We can argue that the length of the suffix is $32 - 29$ or 3. So there are $2^3 = 8$ addresses in this block. If the first address is 205.16.37.24, the last address is 205.16.37.31 ($24 + 7 = 31$).

Note

**A block in classes A, B, and C
can easily be represented in slash
notation as**

A.B.C.D/ n

**where n is
either 8 (class A), 16 (class B), or
24 (class C).**

Example 13

What is the network address if one of the addresses is 167.199.170.82/27?

Solution

The prefix length is 27, which means that we must keep the first 27 bits as is and change the remaining bits (5) to 0s. The 5 bits affect only the last byte. The last byte is 01010010. Changing the last 5 bits to 0s, we get 01000000 or 64. The network address is 167.199.170.64/27.

Example 14

An organization is granted the block 130.34.12.64/26. The organization needs to have four subnets. What are the subnet addresses and the range of addresses for each subnet?

Solution

The suffix length is 6. This means the total number of addresses in the block is 64 (2^6). If we create four subnets, each subnet will have 16 addresses.

Solution (Continued)

Let us first find the subnet prefix (subnet mask). We need four subnets, which means we need to add two more 1s to the site prefix. The subnet prefix is then /28.

Subnet 1: 130.34.12.64/28 to 130.34.12.79/28.

Subnet 2 : 130.34.12.80/28 to 130.34.12.95/28.

Subnet 3: 130.34.12.96/28 to 130.34.12.111/28.

Subnet 4: 130.34.12.112/28 to 130.34.12.127/28.

See Figure 5.15

Figure 5-15

Example 14

